

Final Report

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Part IIA Project: *Neural Control with Adaptive State Estimation* (GG4)

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Introduction

We are given a `Brain` and a Brain–Machine Interface (BMI), which controls an arm. We are tasked with controlling the position of the hand at the end of the arm. The `Brain` takes $n_u = 2$ inputs and produces $n_y = 16$ outputs; we assume it is a Linear Gaussian State–Space Model (LGSSM). The BMI takes the 16-dimensional observations from the `Brain` and applies a fixed linear decoder to map them to 2 latent states: x_1 , the muscle selection rhythm; and x_2 , power rhythm. The BMI stores a rolling history of x_1 , which is passed through 4 fixed band–pass filters which measure how strongly x_1 oscillates at that muscle’s defined excitation frequency. A history of x_2 is also stored in state, and the current value is compared to a half– and full–period in the past: if strong oscillation is found, evidence accumulates; otherwise, evidence decays, producing a smooth power state. Muscle activation is the product of activation and the selection vector. Each muscle produces force proportional to its activation, and for each joint, the net force is the sum of signed muscle forces. Positive forces represent flexion; negative forces represent extension. A dry-friction threshold must be overcome, and once this is overcome, the remaining net force (once subtracting the dry friction) is divided by wet friction representing viscous resistance. Joint angles are integrated, and 2-link arm kinematics give the hand position.

Rules of engagement

The holistic view of this project is to be able to reason and interpret a controller which takes observations from a brain (e.g. via electrodes on the scalp) to move a prosthetic arm. Critically, the brain is a black box and its model “parameters” cannot be directly observed. In the spirit of this view, we propose specific rules of engagement surrounding our analysis. We do not attempt to decompile the provided `Brain` executables to derive the system matrices. We do not average over multiple `Brain` seeds, removing noise to derive system matrices, we reserve this for evaluation purposes only to demonstrate robustness against noise. Where we do run experiments that are not physically realisable, we do this to demonstrate the direction of our analysis, and we replicate results in empirically valid ways alongside. We do not directly use the weights of the BMI in our control efforts: we allow ourselves to read their values for experimental design, but any results we must prove empirically before use. We only allow the use of the 16-dimensional

observations from the `Brain` and the hand position given by the BMI for closed-loop control; we do not rely on the internal state of the BMI (e.g., arm angles or latent state variables) for feedback. We also do not expect or attempt to control the elbow joint past 180° to mimic the capabilities of an anatomically accurate arm. We motivate these constraints as to achieve resilience and independence to both: inter-individual variation in brains (and therefore system matrices); and manufacturers of prosthetic arms (and therefore BMI decoding, state, and arm kinematics defined by model weights).

System identification

As we assume the `Brain` is a LGSSM, we explore system identification as a means to understand the system matrices defining the model:

$$\begin{aligned} x_{t+1} &= \underline{A} x_t + \underline{B} u_t + w_t, & w_t &\sim \mathcal{N}(\underline{0}, \underline{Q}) \\ y_t &= \underline{C} x_t + o_t, & o_t &\sim \mathcal{N}(\underline{0}, \underline{R}) \end{aligned}$$

Identification evaluation

To evaluate our estimates, we have two strategies: the reconstruction of observations and the reconstruction of frequency response. To measure the reconstruction of observations, we use the coefficient of determination defined with the Frobenius norm:

$$R^2 = 1 - \frac{\|Y - \hat{Y}\|_F^2}{\|Y - \bar{Y}\|_F^2}$$

This form is chosen for its effectiveness at aggregating errors across all time points and all observation dimensions. We weight this up against its disadvantages, which include ignorance of observation covariance and temporal correlations. To measure the reconstruction of frequency response, we inspect a frequency map derived analytically from estimates of system matrices, and compare characteristics with empirical results from a single run of the `Brain`. We find the transfer function of the LGSSM, dependent on only the deterministic components (matrices $\underline{A}$, $\underline{B}$, and $\underline{C}$):

$$H(z) = \underline{C} (zI - \underline{A})^{-1} \underline{B}$$

We define an estimate to reconstruct the frequency response well if it can minimise the error between analytic and empirical values of $|H(e^{j\omega})|$ across all neurons and frequencies $[0, \pi]$. Given the emphasis on

rhythm in the BMI we weigh frequency response reconstruction more than observation reconstruction if a compromise is necessary.

Identification methods

We run two Brain instances with and without input to isolate and subtract the noise. We can then calculate the exact Markov parameters, construct Hankel matrices, and form a singular-value spectrum. With this analysis we find σ_i/σ_1 fall after $i = 6$, indicating the true latent dimension number to be $n_x = 6$ (Figure 1). However, this method is not physically realisable: you cannot run the same physiological brain twice with the same random state. This would mean rewinding a brain, re-run a trial without a stimulus, and subtract: noise is truly stochastic outside of a simulation. A physically realisable single-run Ordinary Least Squares (OLS) regression estimate uncovers a noise floor that obscures modes 3–6. Balancing likelihood and model complexity with Bayesian Information Criterion (BIC) model selection, we identify $n_x = 2 \neq 6$ (Figure 1).

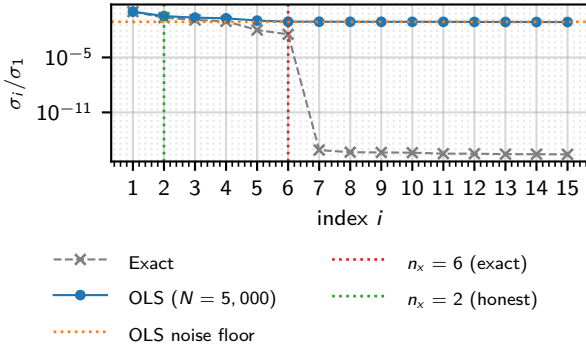


Figure 1: Normalised Hankel singular values σ_i/σ_1 of the 40×60 block Hankel matrix, comparing the exact same-seed-subtraction spectrum (a theoretical reference for the true order) with the physically realisable single-run OLS spectrum. [Source](#)

We plot a heatmap of $|H(e^{j\omega})|$ and find that Canonical Variate Analysis (CVA) and Expectation-Maximisation (EM) to reconstruct u_1 well, but not u_2 over all n_x candidates $\{2, 4, 6\}$ (Figure 4, N.B. only $n_x = 4$ plotted). We turn to an estimation method that uses the Eigensystem Realisation Algorithm (ERA) and EM that utilises OLS-estimated Markov parameters and find similar results for $n_x = 2$: good reconstruction of u_1 but poor for u_2 . However, in contrast to CVA+EM, reconstruction of the u_2 frequency response improved with $n_x = 4$ and $n_x = 6$ (Figure 4). More generally, we observe that the Brain acts as a low-pass filter. The observation reconstruction quality remained similar between CVA+EM and ERA+EM (Figure 2).

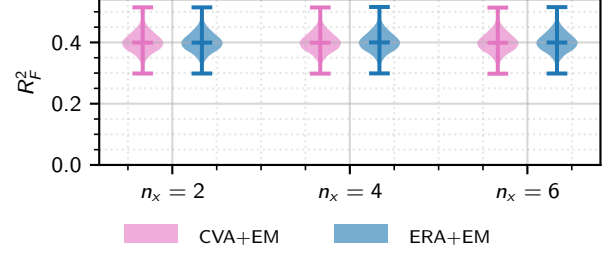


Figure 2: Frobenius R_F^2 of one-step-ahead Kalman predictions across $N = 1000$ held-out trials of $T = 2000$ steps (first 100 steps excluded as Kalman transient). Every estimator and n_x combination clusters at a median $R_F^2 \approx 0.40$. [Source](#)

As there was not a huge difference between $n_x = 4$ and $n_x = 6$, we select ERA+EM and $n_x = 4$, which yields system matrices visualised in Figure 3.

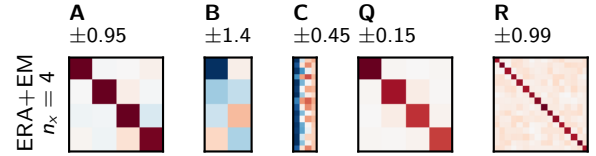


Figure 3: Identified system matrices A, B, C, Q, R of the chosen model. Diverging colour scale is symmetric about zero. [Source](#)

Closed-loop spectral control of brain observations

We test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population mean activity. We motivate the following evaluation framework and method with the emphasis on controlling rhythms in the Brain observations for influence over BMI latent variables and the arm downstream.

Spectral control evaluation

We use the following metrics to evaluate each controller: spectral Root Mean Square (RMS) error, which measures amplitude deviation at target frequencies; the maximum amplitude error, the worst-case per-component deviation; and the off-target power ratio, the mean spectral amplitude at non-target frequencies, relative to mean target amplitude.

Spectral control method

We offer 3 candidates for control strategies: Linear Quadratic Gaussian (LQG) to respect the Brain's LGSSM structure; Linear Quadratic Integral (LQI) to correct steady-state error; and Model Predictive Control (MPC) to optimise for and appreciate constraints on $u_i \in [0, 1]$.

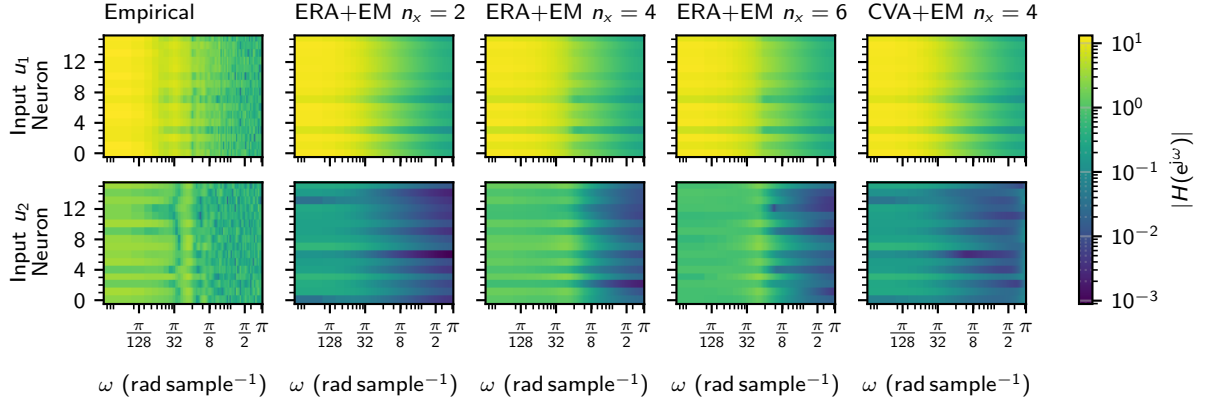


Figure 4: Per-neuron frequency selectivity $|H(e^{j\omega})|$ for inputs u_1 (top row) and u_2 (bottom row), all neurons against $\omega \in [0, \pi]$ on a shared logarithmic colour scale. Leftmost column: empirical response (single-run OLS, 5000 steps, DFT zero-padded to 512 points). Remaining columns: analytic ERA+EM responses for $n_x \in \{2, 4, 6\}$ and CVA+EM at $n_x = 4$ for comparison. [Source](#)

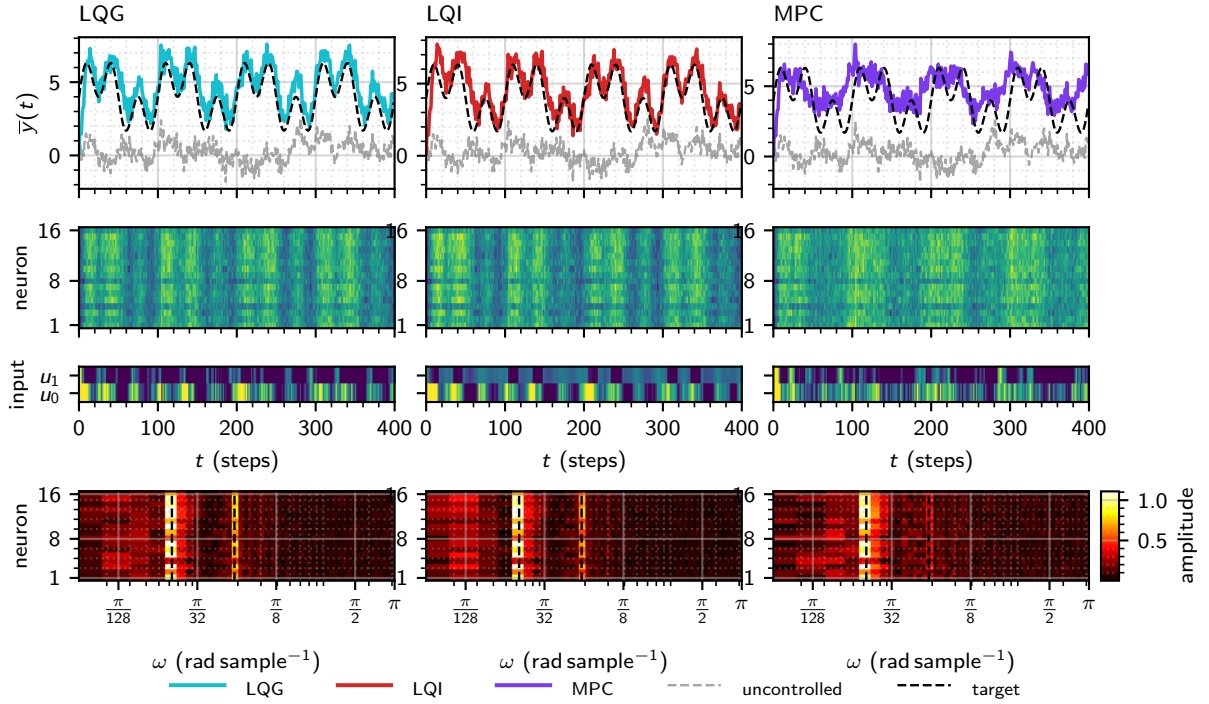


Figure 5: Closed-loop spectral control of the population mean. Target: two sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample (amplitudes $A \in \{1.50, 1.50\}$) about a mean of 4; each controller tuned by random search (8 trials). Columns are LQG, LQI, MPC. Rows, top to bottom: population mean $\bar{y}(t)$ with the uncontrolled baseline and target reference; per-neuron observations $y(t)$; applied inputs $u(t)$; per-neuron amplitude spectra with target frequencies marked. [Source](#)

Spectral control results

Inspecting the raw population mean traces, we find LQG overestimates the population mean with a steady-state error, which is removed effectively by LQI. MPC fails to reconstruct the higher frequency components. We select LQI due to its lower control effort compared to LQG and better reconstruction performance over MPC.

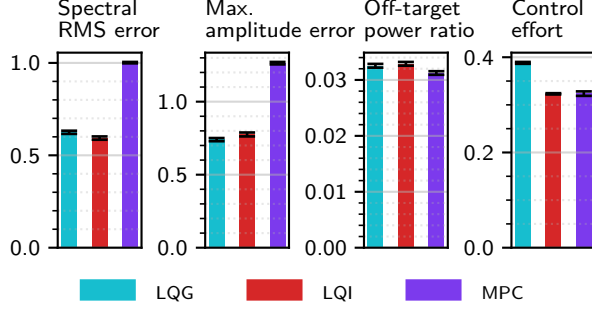


Figure 6: Spectral tracking metrics across $N = 20$ independent Brain seeds, for controllers tuned by random search (8 trials, seed 1) on the target of Figure 5. Bars show the mean; error bars the standard error. Panels left to right: spectral RMS error, maximum amplitude error, off-target power ratio. [Source](#)

1 Hand control

The mechanism of the BMI of muscle selection and a shared power state means that simultaneous control of different muscles is unlikely, as there is no granular control to direct different muscles at different power at the same time. This constrains our control strategy to be a sequential one, directing one muscle at a time.

Evaluating hand control

We aim to minimise the final distance to target, the Euclidean distance between the hand and the target at the end of a trial. We define a success threshold at 10. We aim to maximise the reach rate, the fraction of trials that reach within this success threshold. This metric captures reliability across diverse targets in the reachable area: a controller with low mean final distance but occasional failures is not practically useful for a prosthetic. We minimise the time to threshold, the number of steps until the hand first enters the distance success threshold: a useful prosthetic should converge in reasonable time to provide utility. We maximise the sector fraction, the proportion of time the hand spends in the optimal sector: an angular wedge containing the target and the start position. We minimise the rotational distance, the total angle swept by the hand trajectory over the trial. These two metrics penalise inefficient trajectories (e.g. looping behaviours), which would be inconvenient and unpredictable for a prosthetic. We minimise the con-

trol effort, the sum of square brain inputs, in an effort to make inputs more interpretable and efficient.

Muscle selection

We sweep frequencies for both inputs and measure shoulder and elbow angles, determined by inverse kinematics to find the specific frequencies that excite specific muscles. We empirically recover the frequency bands defined in the BMI weights using a single seed, shown in Table 1 and Figure 7.

Muscle	ω_{true}^* (rad step ⁻¹)	ω_{rec}^* (rad step ⁻¹)	$\Delta\omega$ (rad step ⁻¹)
S+	0.4378	0.4263	-0.0115
S-	0.7854	0.7800	-0.0054
E+	1.2875	1.3006	0.0131
E-	1.9764	1.9458	-0.0306

Table 1: True bandpass-filter centre frequencies ω_{true}^* (from the BMI weights) against the values ω_{rec}^* recovered by the open-loop sweep (Figure 7); $\Delta\omega = \omega_{\text{rec}}^* - \omega_{\text{true}}^*$. Rows are ordered by ascending frequency: shoulder flexion (S+), shoulder extension (S-), elbow flexion (E+), elbow extension (E-). [Source](#)

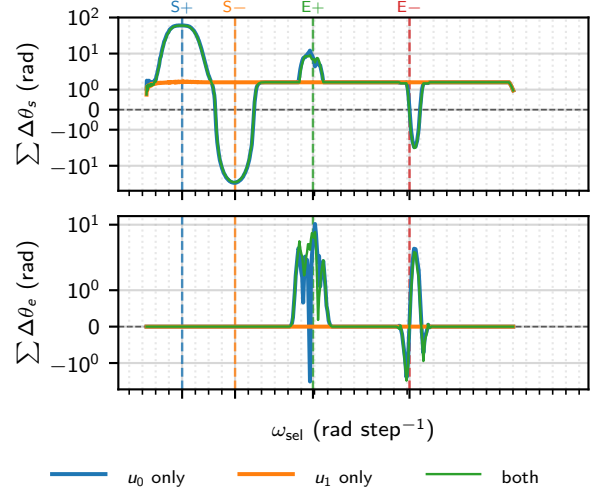


Figure 7: Open-loop sinusoidal drive (amplitude $A = 0.45$ about 0.5) applied to u_0 or u_1 independently at each of 512 log-spaced frequencies $\omega_{\text{sel}} \in [0.188, 2.639]$ rad step⁻¹. Top: signed cumulative shoulder displacement $\sum \Delta\theta_s$; bottom: cumulative elbow displacement $\sum \Delta\theta_e$, each summed over a 900-step post-warmup window (symlog, linear threshold ± 1 rad; 5-sample moving average). Vertical lines mark the four bands, located on the u_0 response by peak displacement (S $\pm$) and peak elbow selectivity $|\Delta\theta_e|/(|\Delta\theta_e| + |\Delta\theta_s|)$ (E $\pm$). [Source](#)

We observe that amplitude decreases with frequency, making it harder to drive muscles that require higher frequencies. This may be due to the low-pass action of the Brain. We also observe cross-talk for elbow movements but not shoulder movements: the

shoulder can be moved independently of the elbow, but the elbow cannot be moved independently of the shoulder.

We then ask whether neural observations carry a muscle-specific spatial code: if different neurons respond preferentially at each selection frequency, the control target could be weighted to favour one muscle over another. For each selection frequency we record the full $n_y = 16$ D amplitude spectrum and normalise the per-neuron profile by its own neuron-mean (Figure 8).

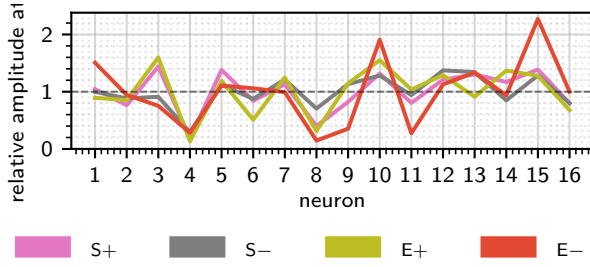


Figure 8: Neural correlates of muscle selection. For each of the four selection frequencies identified in Figure 7, a sustained open-loop sinusoidal trial ($u_0 = 0.5 + 0.4 \sin(2\pi f^* t)$, $u_1 = 0.5$) recorded $n_y = 16$ D neural observations over 1024 steps following 500 burn-in steps. Each line shows the amplitude at ω_{sel}^* per neuron, normalised by the muscle's own neuron-mean; the dashed line marks the mean ($= 1$). Profiles that collapse onto each other confirm that every muscle recruits the same relative neural activation pattern, so the $16 \text{ D} \rightarrow x_1$ projection is not muscle-specific. [Source](#)

Muscle identity is therefore encoded temporally by the oscillation frequency of x_1 , not spatially by which neurons are active (Figure 4). No differential weighting of neurones in the spectral control target offers muscle selectivity beyond what the BMI's bandpass filters already provide. This justifies controlling the population mean at the spectral target.

Control method

The core pipeline is a 3 step process. First is band-identification, an open-loop sinusoidal sweep over both brain input channels, with cumulative shoulder and elbow displacement (inferred via inverse kinematics) recorded per frequency. The four bands are identified, which correspond to S+, S-, E+, and E-. The next step is gain calibration, where a short-closed loop LQI trial at each band frequency targets a sinusoidal spectral setpoint, and mean joint angle per unit amplitude forms a 2×4 gain matrix where rows are joints, and columns are bands. The third is the main control loop, where inverse kinematics converts cartesian target to a joint-angle goals, band amplitudes are solved from the gain matrix and an LQI controller optionally closes the loop on the population mean.

The elbow to shoulder cross-talk motivates a con-

trol strategy that first achieves the correct elbow angle, then reduces control gain over the elbow, and focuses on using S+ and S- to correct the shoulder angle. In addition, during the initial correction angle phase, we should alternate between correcting the elbow angle and correcting the shoulder angle, as attempting to move the elbow moves the shoulder. To mitigate cross-talk further, we drive the shoulder in the opposite direction of the expected cross-talk.

We always begin with targeting elbow error first. Each elbow/shoulder phase has a minimum length and maximum length. In addition, when controlling the elbow, there is a maximum shoulder drift, such that if upon moving the elbow the shoulder drifts too much, we switch to controlling the shoulder. We also provide per-band gain multipliers applied to the gain matrix values before the solve, to further tune individual band aggressiveness. Once the elbow angle is below threshold, we begin damping the shoulder velocity in an effort to reduce oscillations.

Hyperparameter optimisation

With so many model parameters defined by the complex sequential controller design, we turn to hyperparameter optimisation to search the parameter space for an optimal solution. We use Tree-structured Parzen Estimator (TPE) via the optuna library to run two phases: first, a large set of Bayesian Optimisation (BO) trials; then the top trials are re-evaluated on a full target grid, and the best is selected. We define an objective score:

$$S = \bar{d}_{\text{final}} - 20 \bar{f}_{\text{sector}} - 10 \bar{p}_{\text{reached}} + 50 \overline{\max(0, \theta_{\text{rot}} - 2\pi)}$$

This directly penalises final distance, inefficient trajectories, and rewards successful reach rate to create a more interpretable optimisation process. Figure 9 shows how the optimisation process did not reduce final distance or the time to threshold, but instead made the trajectory more sensible, increasing the sector fraction and reducing the rotation.

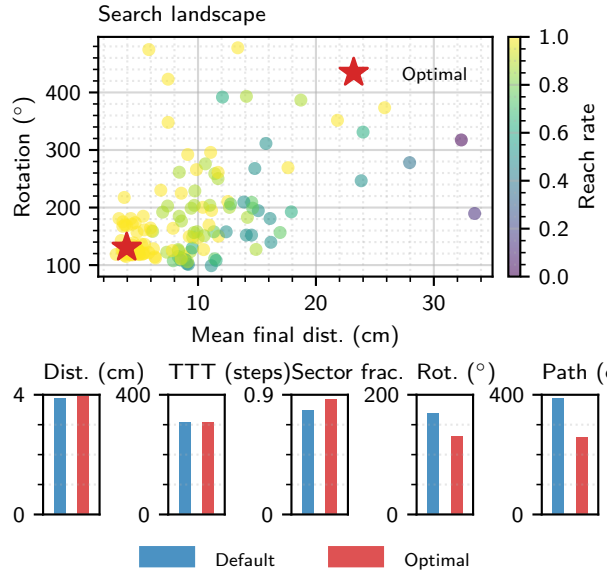


Figure 9: Two-phase TPE optimisation over eight controller parameters (150 trials). Left: per-trial objective score (faint points) and running best (line). Centre: search landscape, each trial placed at its mean final distance and sector fraction, coloured by reach rate, with the selected configuration starred. Right: default versus optimised controller across four metrics on the full evaluation grid (24 targets, 3×8 polar, $r \in [20, 55]$ cm). [Source](#)

Results

The selected configuration and the range searched for each parameter are given in [Table 2](#); no optimum sits on a search bound. The most optimal controller's metrics are summarised in [Table 3](#), its losses over time are plotted in [Figure 10](#), an example trajectory is illustrated in [Figure 11](#), and a heatmap of targets and the final distances to them is drawn in [Figure 12](#).

The shoulder bands were scaled up and elbow bands were scaled down: the optimiser independently rediscovered the cross-talk hypothesis, that the elbow is “loud”, driving the shoulder with it, so the controller converges on driving the elbow slowly. In addition, the minimum and maximum phase lengths are short, just a little larger than the state held by BMI, also rediscovering the hypothesis of small elbow movements that require quick and often shoulder corrections to minimise drastic cross-talk effects. The need for shoulder velocity damping is consolidated by the optimal controller converging on a high damping coefficient. The need for the elbow to be held once correct is also confirmed, due to the hard holding coefficient. The benefit of opposite direction shoulder drive in an effort to minimise cross-talk effects didn't seem to be valuable, likely due to the lack of granular control over selected muscle power. The most surprising result is that open-loop control wins over closed-loop control. We argue that the population-mean spectral target is only a loose proxy for the muscle selection latent state.

In addition, as the Brain is a linear system, sinusoidal inputs produce sinusoidal outputs so a BMI based on rhythm can continue to operate without closed-loop control.

Parameter	Range	Optimal
Shoulder band gain	[0.50, 2.00]	1.5
Elbow band gain	[0.10, 1.50]	0.24
Cross-talk pre-drive	[0.05, 0.50]	0.14
Elbow hold once placed	[0.10, 0.80]	0.62
Shoulder velocity damping	[2.0, 50.0]	30.3
Min. phase length	[10, 60]	30
Max. phase length	[50, 200]	78
Close loop on pop. mean	{F, T}	F

Table 2: Optimal Phase-B controller hyperparameters from the TPE search (150 trials), shown with each parameter's search range. [Source](#)

Metric	Default (mean $\pm$ SD)	Optimal (mean $\pm$ SD)
Final dist. (cm)	3.90 $\pm$ 2.76	3.95 $\pm$ 1.42
Reach rate	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
TTT (steps)	308 $\pm$ 155	309 $\pm$ 147
Sector frac.	0.782 $\pm$ 0.126	0.871 $\pm$ 0.082
Rot. dist. (°)	169.3 $\pm$ 121.8	130.6 $\pm$ 65.5
Path length (cm)	389.4 $\pm$ 211.3	259.6 $\pm$ 68.4

Table 3: Mean $\pm$ SD of each metric across the 24-target grid for the default and Bayesian-optimised controllers (TTT, time to threshold). [Source](#)

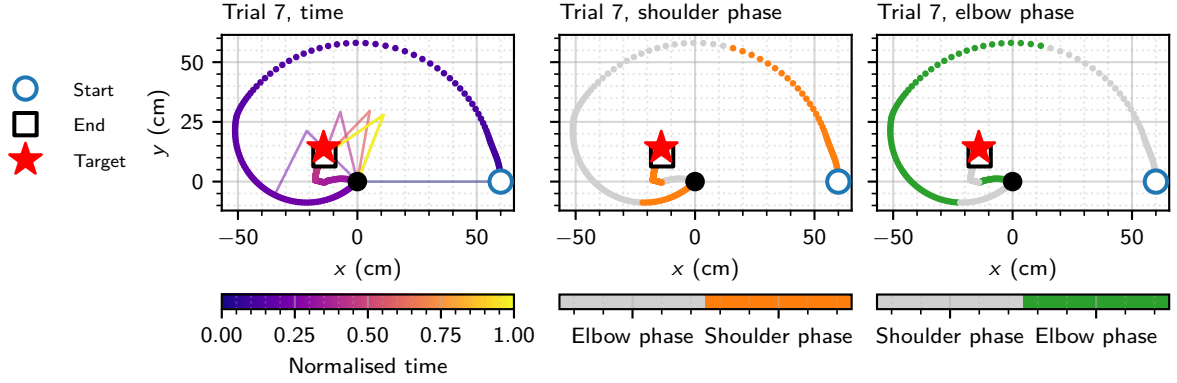


Figure 11: Hand trajectory for demo trial 7 (seed 0, $T = 1000$ steps). Left: path coloured by time, with 6 arm-posture snapshots at evenly spaced steps. Centre and right: the same path coloured by control phase, highlighting the shoulder- and elbow-correction phases respectively. Final distance to target 3.2 cm. [Source](#)

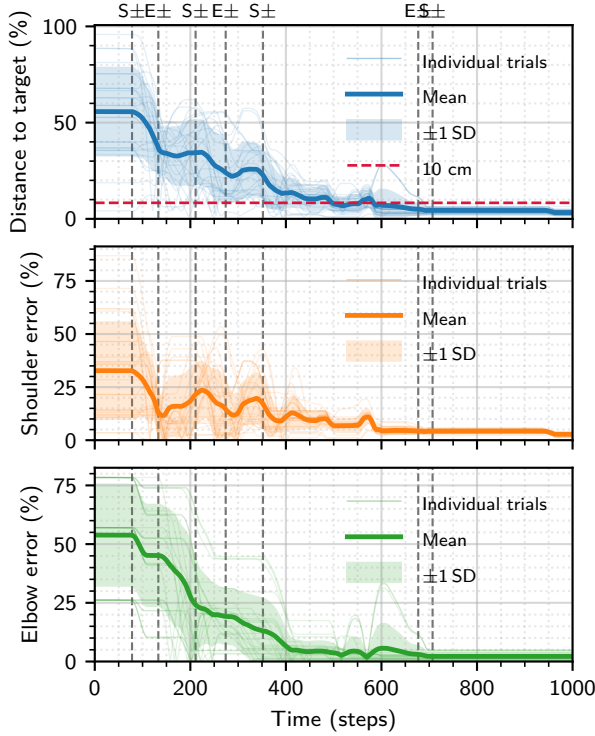


Figure 10: Error time series for the sequential open-loop controller on the 24-target grid (seed 0, $T = 1000$ steps). Top: distance to target as a percentage of the 120 cm maximum reach. Centre and bottom: shoulder and elbow joint error as a percentage of the 180° maximum wrapped error. Vertical dashed lines mark phase switches in demo trial 7, labelled by the activated phase (S $\pm$ shoulder, E $\pm$ elbow). 24/24 trials reached the 10 cm threshold (mean 309 steps); mean final distance 4.0 cm. [Source](#)

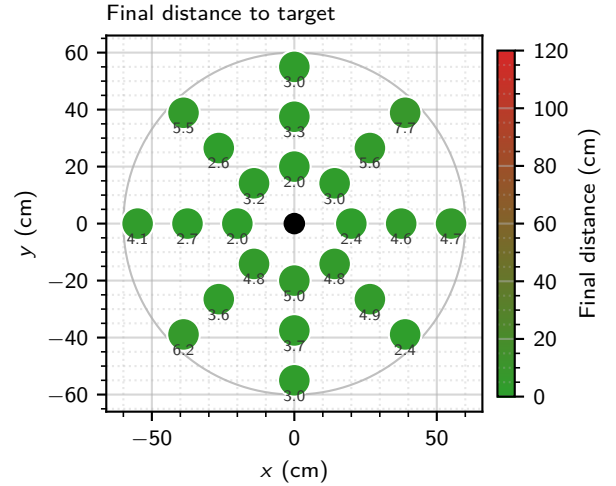


Figure 12: Final distance to target at each location of the 24-target grid (seed 0). The faint circle marks the arm reach boundary ($2L$, twice the link length). [Source](#)

Sensitivity to model parameters

We carry out ablation analysis, varying one parameter of the controller in isolation and holding everything else fixed to measure how sensitive performance is to that parameter. Hyperparameters under analysis include: the elbow band scale, how much the elbow gain is turned down relative to the shoulder gain [Figure 13](#); the blend factor, how much we compensate for cross-talk by driving the shoulder in the opposite direction ??; closed-loop vs open-loop control [Figure 14](#); and whether we constrain the elbow to a maximum of 180° or not [Figure 15](#). We filter the metrics using a one-way Analysis of Variance (ANOVA), run independently for

each metric across all conditions, with a null hypothesis that all group means are equal. A metric is plotted only if the ANOVA p -value satisfies $p \leq 0.05$.

Only the effort significantly varied with the elbow band scale (Figure 13). We find that open-loop control reduced the final distance, had less control effort, and slightly improved the sector fraction Figure 14. We find constraining the elbow to not extend past 180° worsens final distance, improves sector fraction, improves time-to-threshold, reduces effort, reduces rotation and improves the path. The constrained elbow is a much better controller in the areas reachable by an anatomically correct quadrant. However, in regions hard to access by such an arm (i.e., behind and towards the centre of the body, or negative-negative quadrant), its performance is poor, just like a real human.

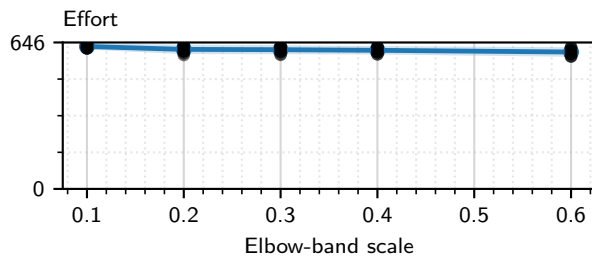


Figure 13: Ablation over the $E \pm$ band-scale gain $\in \{0.1, 0.2, 0.3, 0.4, 0.6\}$ ($S \pm$ fixed at 1.0) on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Lines and shading show the across-seed mean ± 1 SD; dots show per-seed means. [Source](#)

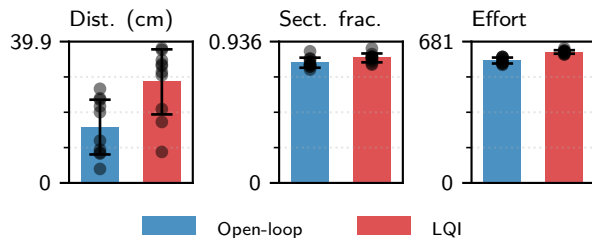


Figure 14: Open-loop (sinusoidal drive) versus closed-loop LQI control on 24 targets across $N = 10$ Brain seeds. Bars show the across-seed mean; error bars ± 1 SD; dots per-seed means. Open-loop final distance 15.8 ± 7.7 cm; LQI 28.5 ± 9.2 cm. [Source](#)

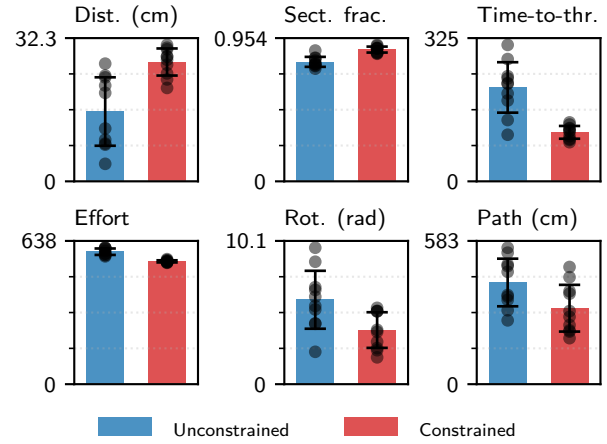


Figure 15: Unconstrained versus anatomically constrained elbow ($\theta_{el} \in [0, \pi]$); setup and encoding as Figure 14. Unconstrained final distance 15.8 ± 7.7 cm (elbow error 29.0°); constrained 26.9 ± 3.1 cm (elbow error 5.3°). [Source](#)

Robustness to noise

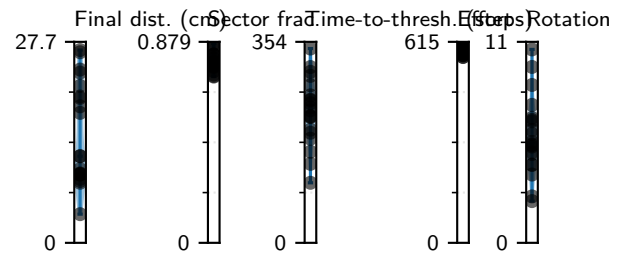


Figure 16: Robustness of the optimal open-loop sequential controller across 20 independent Brain seeds (24 targets per seed, $T = 1000$ steps). Left column: per-seed mean distributions for four metrics; violin bodies show the kernel-density estimate, dots show individual seeds. Right: grand-mean error time series; faint lines show individual-seed per-trial means, bold line and shading show grand mean ± 1 SD across seeds. Top right: distance to target. Centre right: shoulder joint error. Bottom right: elbow joint error. Mean final distance: 15.0 ± 6.9 cm; sector fraction: 0.793 ± 0.041 ; mean reach rate: 0.89. [Source](#)

Limitations and comparison to group solutions

Conclusion

Acknowledgements